An Analytical Battery State of Health Estimation Method

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Abstract— **Decrease in state of health (SoH) of a battery is a measure of end of battery life. Also available battery capacity is dependent on battery life either. Thus accurate estimation of SoH and cycle number of a battery are very important. In this study a method was proposed in order to estimate battery cycle number using ECE 15 driving cycle. Also another method to obtain SoH of a battery using the cycle number is introduced. The results of both methods are compared with the outputs of an experimental setup.**

Keywords; lithium-ion battery, battery model, state of health (SoH), cycle number,

I. INTRODUCTION

Lithium batteries have higher energy densities and open circuit voltages with lower self discharge rates when compared to other secondary battery types [1-3]. Because of these advantages this battery type is commonly preferred in applications. In battery powered systems accurate modeling of batteries has especially significant importance on system analysis. In literature there are many methods or models which can be classified under titles of electrochemical [4-7], statistical [8, 9], equivalent circuit [10-14], and analytical battery models [15, 16]. Also, there are some models named as kinetic battery models which can be categorized under analytical models [17, 18].

Accurate determination of remaining battery life is very important for reliable operation of the system. Thus, a parameter called state of health (SoH) is described. SoH is defined as a figure of merit of the condition of a battery compared to its initial conditions. Unit of SoH is percent points where 100% indicates the initial conditions of a battery. Besides, the usable capacity of a battery depends on its cycle count. The usable capacity of a battery decreases when the its cycle number increases. For instance a curve of relative capacity versus cycle number of the battery, Kokam SLBP55205130H which is used in this study is given in Fig. 1. From figure it can be noticed that a battery's usable capacity changes with cycle number. Also the red line in the figure indicates that a battery is unavailable after its usable capacity drops down under 80%. Therefore the cycle count and SoH are not only used in order to estimate the lifetime of a battery, but also they are important for the estimation of usable capacity amount. Additionally, the cycle number versus capacity information provided by manufacturers are results of aging test series that contain specific charge and discharge cycles. For this reason more accurate methods are needed to estimate SoH under different conditions.

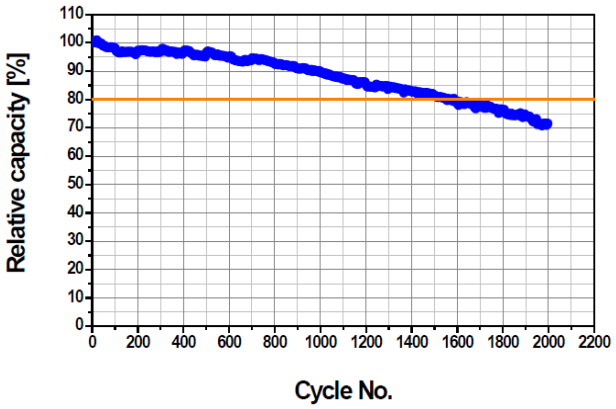


Fig. 1. Relative capacity versus cycle number curve of Kokam SLBP5520510H battery

About SoH here is a wide spectrum of methods presented. [19] focused on using conductor technology together with other observable parameters for telecommunication applications. Also in [20] other alternatives such as electrochemical impedance spectrometry (EIS) or measuring parameters of a battery after discharging it to a fixed cut-off voltage in order to measure SoH are studied. Nevertheless, estimation based methods are more popular. Advanced regression, classification and state estimation algorithms can be used for collecting data for battery health management [21]. Also statistical parametric models are developed in order to estimate error time [22], where in [23] a model was proposed for crank capability prognosis and impedance spectrometry is used. In some studies the focus is directed towards hybrid electrical vehicles where state estimation techniques such as extended Kalman filter are used [24, 25]. Autoregressive integrated moving average (ARIMA) and artificial neural networks are used for both clustering of measured data and estimation [26]. Dynamical models representing nonlinear potentials of lithium-ion batteries such as temperature changes, heat effects, and transient response are generated in [27]. These models are developed by using resistive companion method to avoid electrochemical calculations. In [28] parameters are estimated by using operating current and voltage quantities in an electrical battery model. In order to attain a dynamic estimation, an adaptive observer is designed and Lyapunov stability theorem is used to ensure observers convergence. In [29] pattern recognition is used to monitor health of batteries in a stack. However, when the environmental conditions and load

variations are considered, obtaining or estimating the remaining battery life is difficult. A Support Vector Machine (SVM) method is introduced in [30], a Bayesian implementation of generalized linear form of SVM, Relevance Vector Machine is studied in [31], and RVM is used with Particle Filter algorithms in [32].

In this study a method for SoH and cycle number estimation of a Lithium-Ion battery during a driving cycle is proposed where ECE15 is accepted as the reference driving cycle. The method is based on estimating cycle number and, calculating SoH with the help of correlation between cycle number and SoH. A battery model that was proposed in [33] is used for the method and the results are verified by using experimental ECE15 tests.

In the following section the basis of experimental setup is introduced and the equipments are listed.

In the third section cycle number and SoH estimation methods are described.

In the latter sections the results of the methods were discussed and compared with the outputs of experiments.

II. Experimental Setup

In this study, the objective of the experimental setup is rapid aging of batteries and acquisition of parameters such as voltage, current, and temperature. The setup consists of a programmable DC power supply, Chroma 62050P, a DC electronic load, Chroma 63297, data acquisition cards, NI 9206 and NI 9213, and a system design software, Labview. The scheme of the setup is demonstrated in Fig.2.

The setup was used to perform the test procedures which are required to generate the battery model proposed in [33]. In this study, the experimental setup is not only used to generate battery model, but also used for aging experiments and to get outputs of ECE15 profiles of battery in different cycle numbers. The experimental setup is shown in Fig. 3.

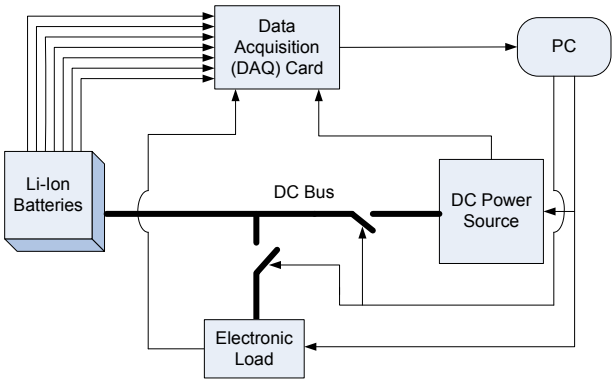


Fig. 2. Schematic of experimental setup.

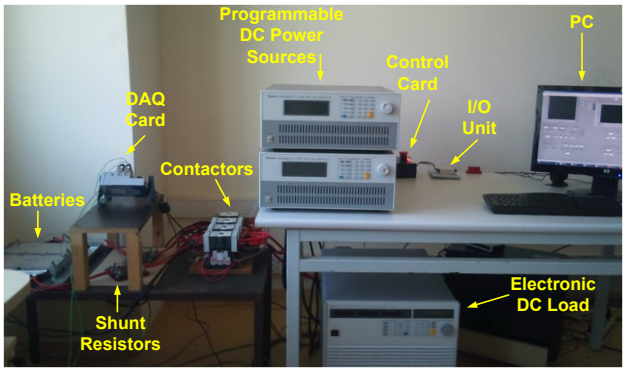


Fig. 3. Experimental setup.

III. Proposed Method

The proposed method uses the cycle number to estimate the SoH of a battery. Thus, it is necessary to determine the cycle number at the first step. One of the main objectives of the study is to achieve cycle number and the SoH of a battery during a drive. Hence, ECE 15 Urban Driving Cycle is used for this purpose. ECE 15 is a driving cycle which represents the driving parameters of an electric vehicle (power, current or speed) versus time. ECE15 driving cycle indicating drawn current is shown in Fig. 4.

Aging of a battery is a long process. Thus, behaviors and responses to ECE 15 of batteries at different ages are simulated by using the model instead of experiments. The model is an equivalent circuit model where the open circuit voltage is calculated by an analytical expression [33].

A. Estimation of Cycle Number

A response of a battery to ECE 15 cycle has some significant sections that can be used to distinguish the curves of different cycle numbered batteries. In the Fig. 5, the ECE 15 simulation output of a battery in different cycles is given. Fig. 5 shows battery voltages and three downward slopes. Bat.@100 and Bat.@700 are the voltages of the battery in 100th and 700th cycles, respectively.

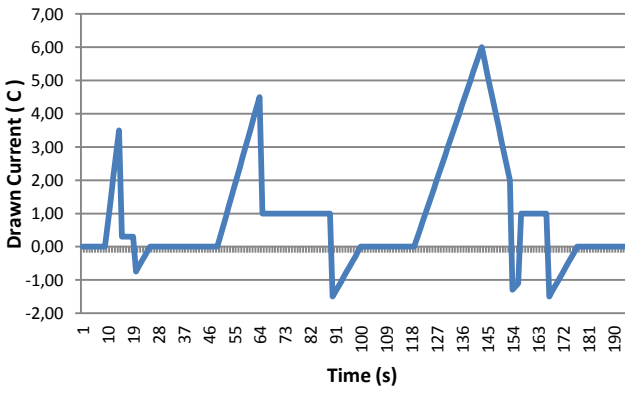


Fig. 4. ECE 15 Urban Driving Cycle.

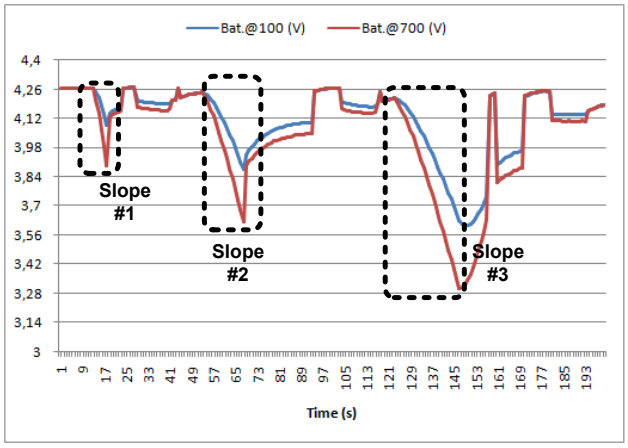


Fig. 5. ECE 15 simulation outputs.

According to the figure, one can comprehend that the slope angles change in each different cycle. By the help of these changes a general formulation can be considered to decide cycle number. A linear equation of a downward slope can be formulated as:

$$y = \frac{a}{x + b} \tag{1}$$

where a and b are parameters and x and y are the corresponding values of points of a slope in time and voltage axes respectively.

For instance, the linear equations of 100th, 300th and 500th cycles of a battery can be obtained for the third slope as below.

$$f_3(100) = \frac{49284{,}283}{x + 11832.95} \tag{2}$$

$$f_3(300) = \frac{35101.534}{x + 8503.278} \tag{3}$$

$$f_3(500) = \frac{25176{,}547}{x + 6130{,}155} \tag{4}$$

From above equations a and b coefficients for every cycle can be sorted as $a_{100} = 49284.283$, $a_{300} = 35101.534$ and $a_{500} = 25176.547$ where $b_{100} = 11832.95$, $b_{300} = 8503.278$, and $b_{500} = 6130.155$.

Similar equations for different cycles are achieved by the help of model. Two different general equations are obtained by Lagrange polynomials using a and b coefficients separately. These are :

$$f_3(a) = 42.455\ 10^{-7} a^2 - 0.146\ a + 1237.934 \tag{5}$$

$$f_3(b) = 25.09\ 10^{-8}\ b^2 - 0.035\ b + 1229.097 \tag{6}$$

where, $f_3(a)$ and $f_3(b)$ are the general functions of the third slope depending on coefficients a and b respectively. Similarly for the first and second slopes general equations are obtained as below. This equation are generated by using the a and b values and curve fitting.

$$f_1(a) = 159.321\ 10^{-7} a^2 - 0.171\ a + 1130.544 \tag{7}$$

$$f_1(b) = 95.56\ 10^{-8}\ b^2 - 0.065\ b + 1116.872 \tag{8}$$

$$f_2(a) = 27.072\ 10^{-7}\ a^2 - 0.115\ a + 1051.669 \tag{9}$$

$$\mathrm{f_2(b)} = 17.46\ 10^{-8}\ \mathrm{b^2} - \ 0.029\ \mathrm{b} + 3.065.408 \tag{10}$$

The cycle number can be calculated with the help of above equations by choosing any two points in any defined slope shown in Fig. 5. Coefficients a and b can be calculated from the voltage and time data by using the equations below.

$$a = \frac{v_1 t_1 + v_2 t_2}{(v_1 - v_2)} \tag{11}$$

$$b = v_2 t_2 + v_2 a = v_1 t_1 - v_1 a \tag{12}$$

where, v_1 and v_2 are the voltage values taken at time t_1 and t_2 in a slope.

B. *Estimation of SoH*

Estimation of SoH is different than the estimation of cycle number because SoH is related with some different parameters such as ambient temperature or internal dynamics of batteries, unlikely cycle number. Therefore, it is not possible to calculate SoH with analytical methods. However, in this study an analytical technique is also proposed based on battery specifications supplied by manufacturer.

The graphic which shows relative capacity change by cycle number was given in Fig. 1. After smoothing by Moving Average method, the curve was formulated as an 8th order polynomial by Lagrange polynomials method as:

$$Capacity = 0{,}0096\ C^8 - 0{,}116\ C^7 + 0{,}8643\ C^6 - 3{,}974\ C^5 + \ 10{,}89\ C^4 - 16{,}88\ C^3 + 111.2 \tag{13}$$

where C is the cycle number.

A correlation may be configured between the slopes and the capacity change given in (13). Thereby the percent capacity may be formulated.

$$Capacity\ (\%) = A\frac{b}{a} \tag{14}$$

where b and a may be achieved from equations (11) and (12), and A is defined as aging coefficient. Values for A is A_1=23.971, $A_2 = 23.583$ and, $A_3 = 23.821$ for first, second and, third slopes, respectively.

As it is mentioned before, it is assumed that the battery health is critical when its relative capacity is dropped down under 80%. When the initial and the maximum capacity value is taken as %100,the percent SoH of a battery can be calculated with the equation below:

$$SoH(\%) = (Capacity\ (\%) - 80).5 \tag{15}$$

IV. RESULTS

A. Estimation of Cycle Number

When two points are selected in the third slope of ECE 15 output of a 400 cycle battery, the parameters are calculated as $a=7190,3$ and $b=29580.89$ from (11) and (12). When these parameters are placed in (5) and (6), cycle number of this battery is calculated as 405.146 depending on a parameter and as 404.745 depending on b parameter. Thus the error can be calculated as 1.18 % and 1.28 % respectively. In Table 1 the results of cycle number estimation method for 100th, 300th, and 500th cycles of a battery is shown for both parameters. Also in Table 2, results of the method for 400th cycle for both three slopes are presented.

TABLE 1 RESULTS OF CYCLE NUMBER ESTIMATION METHOD FOR DIFFERENT CYCLES. (A) FOR PARAMETER a, (B) FOR PARAMETER b

(A).

Cycle	Value of parameter (a)	Calculated cycle number	Error (%)
100	11832.95	104.7727	4.77
300	8503.278	303.4294	1.14
500	6130.155	502.4722	0.49

(B)

Cycle	Value of parameter (b)	Calculated cycle number	Error (%)
100	49284.283	113.5683	13.58
300	35101.534	309.6816	3.23
500	25176.547	506.953	1.39

TABLE 2 RESULTS OF CYCLE NUMBER ESTIMATION METHOD FOR A BATTERY OF 400 CYCLES FOR DIFFERENT SLOPES. (A) FOR PARAMETER a, (B) FOR PARAMETER b

(A).

Slope	Value of parameter (a)	Calculated cycle number	Error (%)
1	3381.289	397.681	0.579
2	6660.464	407.083	1.770
3	7190.3	404.745	1.186

(B)

Slope	Value of parameter (b)	Calculated cycle number	Error (%)
1	13775.371	397.044	0.739
2	27874.041	405.324	1.331
3	19580.089	405.146	1.286

B. Estimation of SoH

The results of SoH estimation method may be verified by using discharge time of a battery under constant current. For instance the battery, which was used in this study has 11 Ah capacity. This means that the battery will be fully empty after discharge with a constant, 11 A current for an hour.

The battery was discharged with a constant current of 1 C (11 A) in 100th, 300th and 500th cycles. Discharge times are 3659 s for 100th , 3438 s for 300th and 3474 s for 500th cycles. The corresponding percent capacities are %101.58, %95.5 and %96.5 respectively.

In the Table 3 results which were calculated by proposed method and obtained from aging tests were compared.

TABLE 3 COMPARISON OF RESULTS OF SOH ESTIMATION METHOD WITH AGING TESTS.

Cycle	Capacity (%)		Error (%)
	Estimated	*Measured*	
100	99.194	101.58	2.348
300	98.694	95.50	3.344
500	67.194	96.50	0.719

V. CONCLUSION

In this study an analytical SoH estimation method is proposed. In order to estimate SoH, cycle number of the battery is needed. Therefore another method is developed to determine the cycle number.These two analytical methods are both compatible with ECE 15 Urban Driving Cycle. Consequently both methods are independent from the battery parameters which are difficult to be obtained during drive.

The cycle number estimation method is a geometrical approach to ECE 15 output of a battery. By using a battery model, three different formulations are developed for each of three slopes of driving cycle. These formulations include two parameters (a and b) which can be obtained from slopes. When compared with the test results, the average error in 3rd slope for 100, 300,and 500 cycle battery is 4.1 %. If cycle numbers were counted than the average error would be 6.813 cycles. The average error in both three slopes for 400 cycle battery is 4.5955 cycles or 1.1485 %.

The proposed SoH estimation method is based on cycle number of a battery. The capacity change curve of an aging battery is smoothed by Moving Average method and an 8th order polynomial was obtained by Lagrange polynomials. The capacity function depends on cycle number and it is different for each three slopes. This difference displayed in the function with three coefficients depending on each slope. When compared with the test results, the average error in 3rd slope for 100, 300,and 500 cycle battery is 2.137 %

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